Peter Dietrich
Senior Vice President and Chief Nuclear Officer

DTE Electric Company
6400 N. Dixie Highway, Newport, MI 48166
Tel: 734.586.4153 Fax: 734.586.1431
Email: peter.dietrich@dteenergy.com

May 16, 2024
NRC-24-0035

10 CFR 50.73

U.S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, DC 20555-0001

Fermi 2 Power Plant
NRC Docket No. 50-341
NRC License No. NPF-43

Subject: Licensee Event Report (LER) No. 2024-002

Pursuant to 10 CFR 50.73(a)(2)(i)(B), DTE Electric Company (DTE) is submitting LER No. 2024-002, "Primary Containment Isolation Valve Declared Inoperable for a Period Prohibited by Plant Technical Specifications".

No new commitments are being made in this submittal.

Should you have any questions or require additional information, please contact Mr. Eric Frank, Manager – Nuclear Licensing, at (734) 586-4772.

Sincerely,

Peter Dietrich
Senior Vice President and Chief Nuclear Officer

Enclosure: Licensee Event Report No. 2024-002, "Primary Containment Isolation Valve Declared Inoperable for a Period Prohibited by Plant Technical Specifications"

cc: NRC Project Manager
NRC Resident Office
Regional Administrator, Region III

Enclosure to
NRC-24-0035

Fermi 2 NRC Docket No. 50-341
Operating License No. NPF-43

Licensee Event Report (LER) No. 2024-002
“Primary Containment Isolation Valve Declared Inoperable for a Period Prohibited by Plant Technical Specifications”

NRC FORM 366 (04-02-2024) **U.S. NUCLEAR REGULATORY COMMISSION**

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
Fermi 2	[x] 050 / [ ] 052	00341	1 OF 3

4. Title
Primary Containment Isolation Valve Declared Inoperable for a Period Prohibited by Technical Specifications

5. Event Date			6. LER Number			7. Report Date		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year
03	17	2024	2024	- 002 -	00	05	16	2024

8. Other Facilities Involved

Facility Name		Docket Number
NA	[ ] 050	
NA	[ ] 052	

9. Operating Mode	10. Power Level
1	099

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	[ ] 20.2203(a)(2)(vi)	**10 CFR Part 50**	[ ] 50.73(a)(2)(ii)(A)	[ ] 50.73(a)(2)(viii)(A)	[ ] 73.1200(a)
[ ] 20.2201(b)	[ ] 20.2203(a)(3)(i)	[ ] 50.36(c)(1)(i)(A)	[ ] 50.73(a)(2)(ii)(B)	[ ] 50.73(a)(2)(viii)(B)	[ ] 73.1200(b)
[ ] 20.2201(d)	[ ] 20.2203(a)(3)(ii)	[ ] 50.36(c)(1)(ii)(A)	[ ] 50.73(a)(2)(iii)	[ ] 50.73(a)(2)(ix)(A)	[ ] 73.1200(c)
[ ] 20.2203(a)(1)	[ ] 20.2203(a)(4)	[ ] 50.36(c)(2)	[ ] 50.73(a)(2)(iv)(A)	[ ] 50.73(a)(2)(x)	[ ] 73.1200(d)
[ ] 20.2203(a)(2)(i)	**10 CFR Part 21**	[ ] 50.46(a)(3)(ii)	[ ] 50.73(a)(2)(v)(A)	**10 CFR Part 73**	[ ] 73.1200(e)
[ ] 20.2203(a)(2)(ii)	[ ] 21.2(c)	[ ] 50.69(g)	[ ] 50.73(a)(2)(v)(B)	[ ] 73.77(a)(1)	[ ] 73.1200(f)
[ ] 20.2203(a)(2)(iii)		[ ] 50.73(a)(2)(i)(A)	[ ] 50.73(a)(2)(v)(C)	[ ] 73.77(a)(2)(i)	[ ] 73.1200(g)
[ ] 20.2203(a)(2)(iv)		[x] 50.73(a)(2)(i)(B)	[ ] 50.73(a)(2)(v)(D)	[ ] 73.77(a)(2)(ii)	[ ] 73.1200(h)
[ ] 20.2203(a)(2)(v)		[ ] 50.73(a)(2)(i)(C)	[ ] 50.73(a)(2)(vii)		

[ ] **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Eric Frank - Manager, Nuclear Licensing	734-586-4772

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	BD	ISV	P312	Yes					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
[x] No	[ ] Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

At 1718 EDT on March 17, 2024, it was identified that E4150-F042, HPCI Booster Pump Suction from Suppression Pool Inboard Iso MOV, was not fully closing without holding down the pushbutton in the Main Control Room. The Primary Containment Isolation function (PCIV) of E4150-F042 was declared inoperable. This condition did not challenge the ability of Primary Containment to perform its overall safety function. On March 19th, actions were taken to replace the associated Closed Auxiliary Seal-in contact block and restore the PCIV function of E4150-F042. A Past Operability evaluation was conducted, which determined that upon a HPCI System steam line isolation signal from the HPCI control logic, E4150-F042 would not fully close. The PCIV function of E4150-F042 was declared inoperable for a period exceeding that which is allowable per TS 3.6.1.3. This condition is being reported under 10CFR50.73(a)(2)(i)(B), operation or condition which is prohibited by plant technical specifications.

There were no safety consequences or radiological releases associated with this event. The cause of the E4150-F042 not fully closing is due to the Closed Auxiliary Seal-in Contact failing to keep the close contactor coil energized due to high resistance across the electrical contacts.

LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Fermi 2	☒ 050 / ☐ 052	00341	2024	002	00

NARRATIVE

INITIAL PLANT CONDITIONS
Mode – 1
Reactor Power – 099

There were no structures, systems, or components (SSCs) that were inoperable at the start of this event that contributed to this event.

DESCRIPTION OF THE EVENT
At 1718 (EDT) on March 17, 2024, it was identified that E4150-F042, High-Pressure Coolant Injection (HPCI) [BJ] Booster Pump Suction from Suppression Pool Inboard Isolation [ISV] Motor Operated Valve (MOV) (Manufacturer Powell Co., Model 3023WE), was not fully closing without holding down the pushbutton in the Main Control Room (MCR). The Primary Containment Isolation function (PCIV) [BD] of the E4150-F042 was declared inoperable. This condition did not challenge the ability of Primary Containment [JM] to perform its safety function as containment integrity would have been maintained by the redundant water seal and by E4150-F041, HPCI Booster Pump Suction from Suppression Pool Outboard Isolation MOV. This condition also did not challenge the ability of E4150-F042 to automatically open when required, therefore the High-Pressure Coolant Injection (HPCI) system remained operable.

This condition was first identified in Fermi 2 condition report (CR) 2023-30437 (05/23/2023) and again in CR 2024-36518 (02/25/2024). Both earlier CRs drew the same inaccurate operability conclusion that the manual seal-in contact via the open pushbutton is not used when E4150-F042 is closed due to a containment isolation signal. Due to this inaccurate operability conclusion in those 2 CRs, a past operability and reportability review were not conducted (action in CR 2024-36518 to understand the inaccurate operability conclusions). During a pre-refueling outage (RF22) review of Main Control Room Deficiency work orders, a DTE Electrician generated CR 2024-36970 (03/17/2024) to review the associated information with E4150-F042. The operability determination with that CR placed the PCIV function of E4150-F042 inoperable. Operations entered Limiting Condition for Operation (LCO) 2024-0087 on 03/17/2024 at 2345 with an expiration date of 03/20/2024 at 2345 to isolate the affected flow path and align Emergency Core Cooling System operations and instrumentation per TS 3.6.1.3, TS 3.5.1, and TS 3.3.5.1. Those actions were completed and LCO 2024-0087 was exited on 03/18/2024 at 0851. At that time operations moved to revised LCO 2024-0087B with an expiration date of 04/01/2024 at 0851 to support the necessary repairs per TS 3.5.1 and TS 3.3.5.1. LCO 2024-0087B was exited on 03/19/2024 at 1039 when repairs were made and the PCIV function of E4150-F042 was restored to operable status.

Based on the past operability review, the time period used to declare the PCIV function of E4150-F042 inoperable is from when the condition was first identified in CR 2023-30437 on 05/23/2023, until the appropriate repairs were made on 03/19/2024.

Per Technical Specification (TS) 3.6.1.3 Primary Containment Isolation Valves (PCIV), each PCIV shall be operable in MODES 1, 2 and 3. CONDITION E states that if any required action and associated completion time of CONDITION A, B, C or D are not met, ACTION E.1 requires the plant to be in MODE 3 in 12 hours, and ACTION E.2 requires the plant to be in MODE 4 in 36 hours.

Immediate troubleshooting was conducted on E4150-F042 failing to fully close. It was determined that E4150-F042 Closed Auxiliary Seal-in contact was intermittently failing to keep the close contactor coil energized due to high resistance across the electrical contacts. This resulted in E4150-F042 failing to fully stroke closed during required surveillances without holding the pushbutton closed in the Main Control Room.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Fermi 2	■ 050 ☐ 052	00341	2024	002	00

NARRATIVE

On 03/19/2024 at 1039, the entire Closed Auxiliary Seal-in contact block was replaced. Following required testing, E4150-F042 was functioning as designed, and the E4150-F042 PCIV safety function was restored to operable status.

SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS
This Licensee Event Report (LER) is being made pursuant to 10CFR50.73(a)(2)(i)(B) as a condition or operation prohibited by technical specifications. There were no safety consequences or radiological releases associated with this event. At no time during this event was there a potential for endangering the public health and safety.

CAUSE OF THE EVENT
The cause for the failure of the Closed Auxiliary Seal-in contact to keep the close contactor coil energized due to high resistance across the electrical contacts was determined to be “dirty contacts”. The cause of the “dirty contacts” is unknown.

CORRECTIVE ACTIONS.
The entire Closed Auxiliary Seal-In contact block was replaced.

PREVIOUS OCCURRENCES
None